

Core Architecture Concepts

- **File I/O (Input/Output):** The process by which a software application interacts with the local storage system to read data or write data to physical files on a hard drive.
- **Context Managers (`with open`):** An architectural safety standard in Python that encapsulates file access. It guarantees that file streams are cleanly terminated and system memory locks are released, eliminating risk of memory leaks.
- **Defensive Programming (`try-except`):** A design philosophy where software anticipates potential runtime failures (like missing files or network drops) and establishes execution fallbacks to prevent the application from crashing.
- **Scope Guarding (`if __name__ == "__main__":`):** A structural checkpoint that detects whether a file is being executed directly by a user or imported as a background library by an orchestrator, preventing execution collision.